Easy Question
while loop and do while loop
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While loop

1:- Sum of Natural Numbers

A teacher asks students to calculate the sum of the first \(n\) natural numbers. Write a Java program that inputs n and calculates the sum using a while loop.

Example: If n = 5, the output should be Sum = 15.

answer:

```
import java.util.Scanner;


public class Main {
 public static void main(String[]args) {
 Scanner input = new Scanner(System.in);
 System.out.println("Enter n");
 int n = input.nextInt();
 int i = 1;
 int sum = 0;
 while (i<=n) {
 sum= sum +i;
 i++;
 }
 System.out.println("sum=" +sum);
 }
}
```

2:- ATM PIN Verification

An ATM asks a user to enter a PIN. The user has a maximum of 3 attempts to enter the correct PIN. Write a Java program using a while loop that

allows the user to try again until the PIN is correct or all attempts are used.

Answer:

```
import java.util.Scanner;


public class Main {
 public static void main(String[]args) {
 Scanner input = new Scanner(System.in);
 int correctPin = 1333;
 int Pin = 0;
 int attempts=0;
 while (Pin !=correctPin && attempts<3) {
 System.out.println("Enter the Pin:");

 int pin = input.nextInt();
 attempts++;
 if (Pin !=correctPin && attempts<3) {
 System.out.println("please Try again");
 }
 }
 if (Pin == correctPin) {
 System.out.println("correct Pin. access granted.");
 }
 else{
 System.out.println("incorrect Pin. ATM blocked.");
 }

 }
}
```

in the first the massage will be : Enter the Pin.
If enter an incorrect Pin the massage will be: please try again Enter the Pin.

Do while loop

1: Number Guessing Game
A program stores a secret number, such as 7. Write a Java program using a do-while loop that repeatedly asks the user to guess the number until the correct answer is entered.
Display:
Too low! if the guess is less than the secret number.

Too high! if the guess is greater than the secret number.
Correct guess! when the user guesses correctly.

Answer:

```
import java.util.Scanner;


public class Main {
 public static void main(String[]args) {
 Scanner input = new Scanner(System.in);
 int secret = 7;
 int guess;
 do {
 System.out.println("guess the number!");
 guess = input.nextInt();
 if (guess < secret) {
 System.out.println("Too low!");
 }
 else if (guess > secret) {
 System.out.println("Too high!");
 }else{
 System.out.println("correct guess!");
 }
 }
 while (guess !=secret);
 }

 }
```

end!